Journal of Environmental Chemical Engineering 13 (2025) 118256

Contents lists available at ScienceDirect

Journal of Environmental Chemical Engineering

journal homepage: www.elsevier.com/locate/jece

Comparative insight into biomass pretreatment by choline chloride-based deep eutectic solvents in relation to their physicochemical characteristics

Jinwen Guo, Guoce Yu *, Jianlong Wang

Laboratory of Environmental Technology, Institute of Nuclear and New Energy Technology, Tsinghua University, Beijing 100084, China

ARTICLE INFO

Keywords:
Pretreatment
ChCl-based DESs
Kamlet-Taft parameters
Physicochemical properties
Enzymatic digestibility

ABSTRACT

Comparative analysis of choline chloride (ChCl)-based deep eutectic solvents (DESs) with different hydrogen bond donors (HBDs) for lignocellulose pretreatment is fundamental and helpful for the design of an appropriate solvent. In this work, nineteen ChCl-based DESs were prepared, which were categorized into four groups based on their HBDs, i.e. amide-based, polyol-based, acid-based and ternary DESs, and were applied to wheat straw pretreatment. The physicochemical properties and Kamlet-Taft (K-T) solvatochromic parameters of the DESs were determined and compared. The net acidity (α-β) for acid-based DESs far exceeded that for amide-based and polyol-based DESs. Acid-based DESs generally outperformed basic and neutral DESs in delignification and hemicellulose removal. ChCl: formic acid (ChCl:FA) can extract 80.7 % of lignin and 81.8 % of xylan. Delignification and xylan removal were positively correlated with the K-T parameters α and α-β of the DESs, indicating the relationship between the hydrogen bond donating/accepting ability and biomass fractionation. Enzymatic digestibility of the pretreated biomass was predominantly influenced by the HBD type of the DESs. The glucose yields for acid-based DESs were mostly higher than those for amide-based and polyol-based DESs. The highest glucose yield of 78.2 %, nearly fivefold that of raw biomass (16.1 %), was obtained by a ternary DES composed of ChCl, lactic acid and glycerol (ChCl:LA:Gly), suggesting its potential for sugar recovery. Pretreatment by ChCl-based DESs was closely related to the properties of the solvents primarily dependent on the HBDs, where an appropriate modular composition can afford efficient biomass fractionation.

1. Introduction

Efforts are constantly being made to explore alternative renewable energy sources that can replace traditional fossil fuels and simultaneously minimize their negative impact on the environment. Lignocellulosic biomass is potentially the most abundant renewable resource for carbon-based fuels and chemicals [1]. The utilization of waste biomass such as agricultural residues and forestry residues is particularly important from the viewpoint of waste recycling and environmental protection [2,3]. However, lignocellulose mainly composed of cellulose, hemicellulose and lignin, exhibits a complex and recalcitrant structure which makes its fractionation difficult [4,5]. The valorization of lignocellulose via biochemical ways requires overcoming its structural recalcitrance, and this makes pretreatment a prerequisite [6].

Pretreatment facilitates the disruption of the rigid structure of lignocellulose by cleaving intermolecular and intramolecular chemical bonds between cellulose, hemicellulose and lignin [7]. The accessibility of the enzymes to cellulose and hemicellulose is then improved, which contributes to the production of more fermentable sugars [8]. However, routine pretreatment methods, such as acids, alkalis, organosolv, hot water, steam explosion and ionic liquid treatment, exhibit the drawbacks of low efficiency, high energy demand and negative environmental impacts [9,10]. Consequently, it is imperative to develop a sustainable pretreatment strategy with low cost, low energy consumption and low waste [4]. Deep eutectic solvents (DESs) are considered a substitute for ionic liquids and a viable alternative to conventional solvents, which offer a combination of low volatility, high solubility, tunable properties, biocompatibility, biodegradability and economic viability [11,12]. Recently, DESs have been attracting intensive attention in lignocellulose pretreatment because of their strong ability to dismantle the lignocellulosic structure, especially the high selectivity towards lignin solubilization [13–15].

DES is a eutectic mixture of binary or ternary components with a specific ratio, containing at least one hydrogen bond acceptor (HBA) and one hydrogen bond donor (HBD), which are connected by hydrogen bonds [16]. Choline chloride (ChCl) is frequently used as an HBA due to

* Corresponding author.
E-mail address: yugc@tsinghua.edu.cn (G. Yu).

https://doi.org/10.1016/j.jece.2025.118256
Received 6 June 2025; Received in revised form 17 July 2025; Accepted 22 July 2025
Available online 22 July 2025
2213-3437/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

its biocompatibility, low toxicity and cost-effectiveness. ChCl can be synthesized with a 100 % atom economy, or extracted from biomass as a component of B complex vitamins, and it has been used as the feed additive, food-grade ingredient, and even vitamin supplement for humans [17]. ChCl-based DESs are of particular interest and hold promise in lignocellulosic biomass fractionation as one green and efficient solvent [18,19].

The application of ChCl-based DESs with amines, amides, alcohols and carboxylic acids as the HBD to biomass pretreatment has been documented and impressive progress has been made. Experiments on ChCl: urea (ChCl:U) [20], ChCl: imidazole [21] and ChCl: monoethanolamine [22] were conducted and the results proved the effective isolation of cellulose and lignin for further valorization. Pretreatment of corncob by ethylene glycol (EG) and glycerol (Gly) based ChCl-DESs showed excellent lignin extraction and glucose yield of more than 70 % and 85 %, respectively [23]. Poplars pretreated by ChCl:Gly released 90 % of total glucose and showed a high bioethanol yield [24]. Moreover, four kinds of diol (EG, propylene glycol, butanediol (BDO) and pentanediol (PDO)) based DESs were compared and exhibited a notable capacity for poplar fractionation, achieving cellulose extraction and remarkable solubilization of lignin (92–96 %) and hemicellulose (85–89 %) [25]. Acidic DESs composed of formic acid (FA) [26] and oxalic acid (OA) [27] as the HBD enhanced the removal of hemicellulose and lignin and increased the accessibility of cellulose. Some studies compared ChCl-based DESs composed of carboxylic acids and polyols, and concluded that acidic DESs performed better in the efficiency of lignin removal than alkaline and neutral DESs [17]. However, it is reported that alkaline and neutral DESs were able to avoid undesired lignin condensation favorable for further lignin valorization [28] and retain more cellulose macromolecular structures advantageous for saccharification [22]. Recently, ternary DESs have been proven to be effective in lignocellulose fractionation. Lewis acids as an HBD in DESs, such as $FeCl_3$, $AlCl_3$, $CuCl_2$ and $ZnCl_2$, can assist in disrupting the ether and ester bonds between lignin and carbohydrates, thereby enhancing the removal of lignin and enzymatic saccharification of biomass [29,30]. Notably, $FeCl_3$-mediated ternary DESs demonstrated excellent biomass fractionation performance, achieving high saccharification (>90 %) and delignification (~80 %) [31,32]. The ternary DES ChCl: oxalic acid (OA): EG achieved 75.3 % delignification and over 95 % cellulose retention, achieving high enzymatic digestibility while protecting the native structure of lignin [33]. In addition, the incorporation of acetic acid (AA) to ChCl:Gly resulted in an increase in lignin removal rate to 63 % and cellulose retention to 78 % [34].

However, most of the studies regarding ChCl-based DESs have so far focused on pretreatment by one or two certain types of DESs, elaborating on the specific pretreatment characteristics and mechanisms. Little attention has been paid to a comprehensive examination of ChCl-based DESs with various HBDs to explore their roles in biomass pretreatment under similar conditions [35,36]. Particularly, the physicochemical properties of the DESs which are closely related with the HBDs have insufficiently been taken account of, which is of much concern for effective biomass fractionation [15]. In view of their fundamental significance as one typical category of DESs, a comparative investigation of ChCl-based DESs with various HBDs and different physicochemical characteristics in pretreatment is essentially worthwhile, which can be beneficial for clarifying the possible pretreatment mechanisms and designing a potentially competent DES.

In this work, we synthesized nineteen ChCl-based DESs which can be classified into four types according to the HBDs, i.e. amide-based, polyol-based, acid-based and ternary DESs. The physicochemical properties and Kamlet-Taft (K-T) solvatochromic parameters of the DESs were determined. Pretreatment of wheat straw by as-prepared DESs and enzymatic hydrolysis of the pretreated biomass were carried out and compared. The pretreatment efficiency and possible mechanisms were explained in detail with regard to the physicochemical characteristics of the DESs.

2. Materials and methods

2.1. Materials and chemicals

Wheat straw was harvested from Lianyungang, Jiangsu Province, China, which contained 37.7 % glucan, 24.2 % xylan, 2.3 % arabinan, and 20.4 % lignin. Wheat straw was washed with deionized water and dried at 40 ℃ for 1 d, and then pulverized by a milling machine (Yidecheng YDK-200CK) and screened to 40–80 mesh for subsequent pretreatment.

Choline chloride (ChCl, 98 %), urea (U, 99 %), methylurea (MU, 98 %), formamide (FM, 99 %), propanediol (PDO, 99.9 %), butanediol (BDO, 98 %), glycerol (Gly, 99 %), ethylene glycol (EG, 98 %), acetic acid (AA, 36 %), levulinic acid (LevA, 99 %), glutaric acid (GA, 99 %), lactic acid (LA, 90 %), propionic acid (PA, 99.5 %), itaconic acid (IA, 99 %), malic acid (MA, 99.5 %), formic acid (FA, 90 %), $FeCl_3{\cdot}6\ H_2O$ (100 %), glucose (99 %), arabinose (98 %), xylose (98 %), Nile Red (NR, 98.0 %) and 4-Nitroaniline (NH_2, 99 %) were of reagent or analytical grade and purchased from Macklin Biochemical Technology Co., Ltd (Shanghai, China). The cellulase Celic® CTec2 was purchased from Sigma-Aldrich (Shanghai, China).

2.2. DES preparation

A series of DESs were synthesized, in which ChCl was used as the HBA, and different amides, polyols and acids were used as the HBD. For a typical synthesis process, ChCl and an HBD were mixed at a certain molar ratio, and the mixture was stirred at 60 or 80 ℃ at 500 rpm for at least 90 min to ensure sufficient molecular contact. The preparation was considered complete when a transparent and homogeneous solution was obtained without any insoluble substance or any crystallization at ambient temperature. In addition to fifteen binary DESs, three ternary ChCl-based DESs were similarly prepared by choosing a typical amide, polyol or acid as the HBD simultaneously incorporating $FeCl_3$ due to its properties as a Lewis acid. The ternary DES ChCl: lactic acid: glycerol (ChCl:LA:Gly) was synthesized considering the excellent enzymatic hydrolysis performance resulting from the treatment by ChCl:LA and ChCl: Gly. Finally, nineteen DESs were acquired and stored in sealed bottles at 4 ℃ for physicochemical characterization and further pretreatment. The compositions of the as-prepared DESs are listed in Table 1, and the chemical structures of the HBA and HBDs involved are shown in Fig. 1.

Table 1
Compositions of the prepared DESs.

HBA	HBD		Molar Ratio	Abbreviation
Choline chloride	Amide	Formamide	1:4	ChCl:FM
		Urea	1:2	ChCl: U
		Methyl	1:2	ChCl:MU
	Polyol	Ethylene glycol	1:3	ChCl:EG
		Propanediol	1:3	ChCl:PDO
		Butanediol	1:3	ChCl:BDO
		Glycerol	1:2	ChCl:Gly
	Acid	Formic acid	1:2	ChCl:FA
		Acetic acid	1:2	ChCl:AA
		Propionic acid	1:4	ChCl:PA
		Lactic acid	1:2	ChCl:LA
		Malic acid	1:1	ChCl:MA
		Itaconic acid	1:1	ChCl:IA
		Levulinic acid	1:2	ChCl:LevA
		Glutaric acid	1:1	ChCl:GA
	Ternary	Formamide: $FeCl_3$	1:4:0.1	ChCl:FM: $FeCl_3$
		Glycerol: $FeCl_3$	1:2:0.1	ChCl:Gly: $FeCl_3$
		Lactic acid: $FeCl_3$	1:2:0.1	ChCl:LA:$FeCl_3$
		Lactic acid: Glycerol	1:2:2	ChCl:LA:Gly

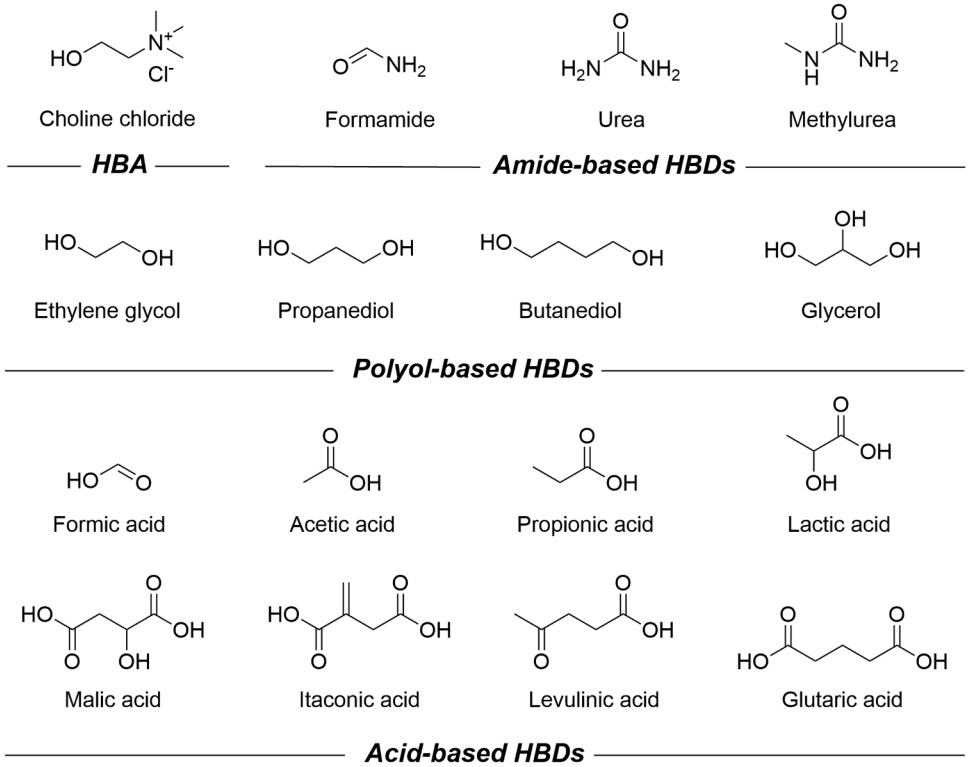


Fig. 1. Chemical structures of HBA and HBDs involved in prepared DESs.

2.3. DES characterization

The physicochemical properties of the prepared DESs were characterized. pH and conductivity of the DESs were assayed by a SevenExcellence multiparameter analyzer (Mettler Toledo). Density was measured by a liquid density meter (Jinheyan JHY-300). Surface tension was determined by the pendant drop method, using a video optical contact angle (Dataphysics OCA 25) with a measuring needle diameter of 1.65 mm. Viscosity was gauged by a rotary rheometer (Anton Paar Physica MCR302), and determined by the measuring cone CP50–0.5 at a shear rate ranging from 1 to 100 s^{-1}. All the parameters were measured at 25 ℃ and data were reported as an average of two measurements.

The Kamlet-Taft solvatochromic parameters of the DESs, including the polarizability (π*), the hydrogen bond donating ability (α), and the hydrogen bond accepting ability (β), were analyzed using the dyes 4-nitroaniline and Nile Red as the solvatochromic probes [37,38]. The dyes were dissolved in methanol at a concentration of 0.001 mol/L, and 200 μL of the solution was transferred to a vacuum drying oven at 40 ℃ to evaporate methanol. 8 mL DES was added and mixed by ultrasonication until the color was uniform. The absorption spectra of the mixture were quantitated by a UV–vis spectrophotometer (Persee T9CS). The solvatochromic parameters were calculated according to the following equations [39]:

$$\pi^* = \frac{19.39 - \nu_{NR}}{2.9912} \tag{1}$$

$$\alpha = \frac{19.9675 - 1.0241\pi^* - \nu_{NR}}{1.6078} \tag{2}$$

$$\beta = 11.134 - \frac{3580}{\lambda_{(NH_2)max}} - 1.125\pi^* \tag{3}$$

$$\nu_{NR} = \frac{10000}{\lambda_{(NR)max}} \tag{4}$$

where $\lambda_{(NR)\max}$ represents the maximum absorption wavelength of Nile Red, and $\lambda_{(NH_2)\max}$ the maximum absorption wavelength of 4-nitroaniline.

2.4. Compositional analysis

The chemical compositions of wheat straw and the DES pretreated biomass were determined according to the standard analytical procedure (NREL/TP-510–42618) of the National Renewable Energy Laboratory [40]. Biomass was hydrolyzed first with 72 % (w/w) sulfuric acid at 30 ℃ for 1 h, and then with 4 % (w/w) sulfuric acid in an autoclave at 121 ℃ for 1 h. The polysaccharide content of the biomass was determined by detecting the sugar concentrations in the hydrolysates with high performance liquid chromatography (HPLC, Agilent 1260), using an Aminex HPX-87H column (300 ×7.8 mm, BioRad, USA) equipped with a refractive index detector. The mobile phase was 5 mM H_2SO_4 with a flow rate of 0.6 mL/min. The column temperature was set at 65 ℃. UV absorption values of the hydrolysates at a wavelength of 320 nm were determined by a UV–Vis spectrophotometer to calculate the content of acid-soluble lignin (ASL). The solid residues were dried at 105 ℃ and then heated in a muffle furnace to determine the content of acid-insoluble lignin (AIL) and ash. The above compositional analysis was performed in duplicate, and the results are expressed as the mean values. Relative standard deviations were within 10 %. The solid yield, delignification and xylan removal were calculated as follows:

$$Solid\ yield(\%) = \frac{Weight\ of\ pretreated\ biomass}{Weight\ of\ untreated\ biomass} \times 100\% \tag{5}$$

$$Deligninfication(\%) = \left(1 - \frac{Lignin\ in\ pretreated\ biomass \times Solid\ yield}{Ligin\ in\ untreated\ biomass}\right) \times 100\% \tag{6}$$

$$Xylan\ removal(\%) = \left(1 - \frac{Xylan\ in\ pretreated\ biomass \times Solid\ yield}{Xylan\ in\ untreated\ biomass}\right) \times 100\% \quad (7)$$

2.5. DES pretreatment of wheat straw

DES pretreatment was performed by mixing wheat straw and DES (mass ratio 1:30) homogeneously in a stoppered conical flask, followed by incubation in an oil bath at 120 ℃ for 3 h. After that, acetone/water (1:1, v/v) was added to the mixture to stop the reaction which was then subjected to vacuum filtration. The solid fraction was washed with acetone/water (1:1, v/v) at least three times until the eluent became colorless. Vacuum filtration against the solid fraction continued until no liquid droplets can be collected. Then the solid fraction was freeze-dried and stored in a desiccator for subsequent analysis. The liquid fraction was subjected to rotary evaporation at 60 ℃ to remove acetone and water from the DES until no droplets were observed in the condensing unit. Thus the DES containing dissolved lignin was collected, from which lignin can be separated and DES recycled.

2.6. Enzymatic hydrolysis of pretreated biomass

Enzymatic hydrolysis was carried out in 50 mM acetate buffer (pH 5.2) at 50 ℃ for 72 h and agitated at 200 rpm in a shaking incubator. A substrate loading of 2 % (w/v) was used to avoid possible substrate inhibition. A cellulase dosage of 15 FPU/g substrate (Novozymes, Cellic® CTec2, 600 FPU/mL) was adopted without adding other β-glucosidases or hemicellulases. Samples were taken at the given time, heated at 100 ℃ for 10 min to terminate the reaction, and filtered through a 0.22 μm membrane. Glucose and xylose in the enzymatic hydrolysate were determined by HPLC as described above. Experiments were performed in duplicate, and results were expressed as the average of the replicates. Enzymatic digestibility was calculated as follows:

$$Glucose\ yield(\%) = \frac{Glucose\ in\ enzymatic\ hydrolysate}{Glucose\ in\ substrate} \times 100\% \quad (8)$$

$$Xylose\ yield(\%) = \frac{Xylose\ in\ enzymatic\ hydrolysate}{Xylose\ in\ substrate} \times 100\% \quad (9)$$

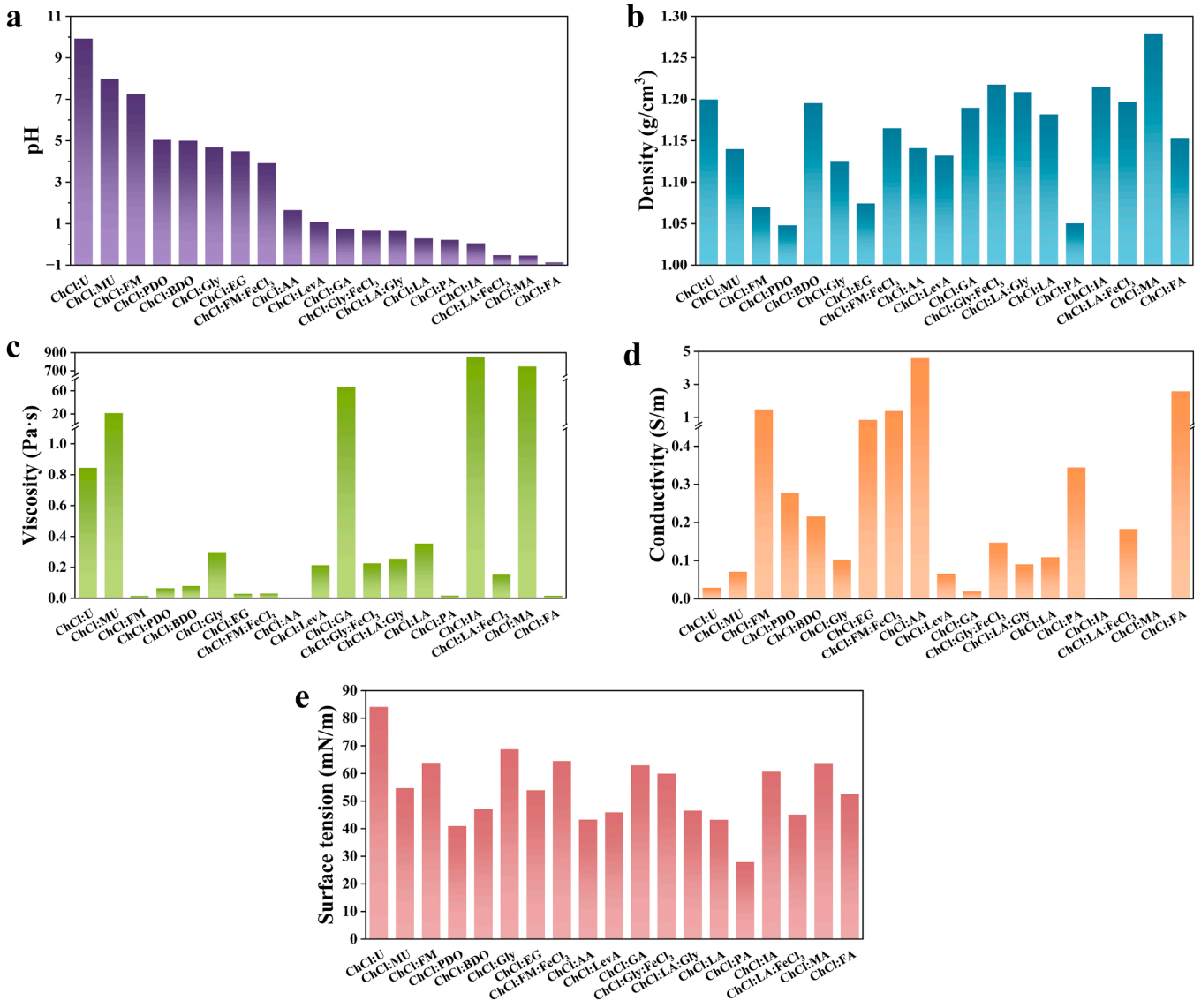


Fig. 2. Physicochemical properties of the prepared DESs. (a) pH, (b) density, (c) viscosity, (d) conductivity, (e) surface tension.

3. Results and discussion

3.1. DES characterization

3.1.1. Physicochemical properties

The physicochemical properties of a DES, which is of modular configuration, are influenced by the structure and the proportion of its components. The ChCl-based DESs prepared here may exhibit unique physicochemical characteristics dominated by their different HBDs. The pH, density, viscosity, conductivity and surface tension of the prepared DESs are shown in Fig. 2.

pH reflected the relative acidity of DESs. It is seen that amide-based DESs were either basic or neutral, with pH of 9.9, 8.0 and 7.2 for ChCl:U, ChCl:MU and ChCl:FM, respectively (Fig. 2a). Polyol-based DESs were weakly acidic and had close pH ranging from 4.5 to 5.0. In contrast, acid-based DESs exhibited strong acidity with pH below 1.7. Notably, ChCl:FA, ChCl:MA and ChCl:LA:$FeCl_3$ had negative pH values in the range of −0.5 to −0.9, exhibiting extraordinary acidity due to their strong acidic HBDs. The pH of $FeCl_3$-mediated ternary DESs was all decreased by introducing the Lewis acid, and interestingly, ChCl:LA:Gly had a pH between the values of ChCl:Gly and ChCl:LA. pH should be closely related to the acidity or alkalinity of the HBDs, which can be displayed in a eutectic mixture.

Density represents the activity and molecular mobility of a solvent. The densities of the prepared DESs were shown to be between 1.0 and 1.3 g/cm^3 at room temperature (Fig. 2b), a range also reported before [41]. ChCl:MA was the densest solvent (1.28 g/cm^3), followed by ChCl:Gly:$FeCl_3$ (1.22 g/cm^3) and ChCl:IA (1.22 g/cm^3), whereas the density of ChCl:PDO and ChCl:PA (about 1.05 g/cm^3) was just slightly greater than water. The disparity in density was considered to result from the different molecular organization or packing of the DESs [42]. The density information may be helpful for considering DESs in biomass treatment and various industrial applications [43].

Viscosity describes the fluidity of a solvent and its resistance to flowing. The viscosity of the prepared DESs varied dramatically as illustrated in Fig. 2c. ChCl:IA and ChCl:MA were highly viscous and possessed a viscosity of over 700 Pa·s, ChCl:GA and ChCl:MU had a viscosity of 66.7 and 21.5 Pa·s, respectively, and ChCl:AA exhibited a viscosity (0.003 Pa·s) very close to water (about 0.001 Pa·s). The mobility of the ternary DES ChCl:LA:Gly (0.26 Pa·s) was found stronger compared with the corresponding binary DESs ChCl:Gly (0.30 Pa·s) and ChCl:LA (0.35 Pa·s). Viscosity revealed the internal resistance of a DES to the flow, and depended on the hydrogen bonding network and the intermolecular interactions between the HBA and HBD [44]. It was thought that the molecular structure (e.g. the alkyl chain length of an HBD) and dimensions of HBDs affected the viscosity of the prepared DESs [42,45]. High viscosity implied low fluidity which would impede the efficiency of mass and heat transfer [46] and cause difficulty in pretreatment operations like solid-liquid separation.

Conductivity represents the ability of ions to flow in a medium. As shown in Fig. 2d, ChCl:AA, ChCl:FA, ChCl:FM and ChCl:EG showed high conductivity of 1.3–4.6 S/m, which correlated with their low viscosity (below 0.03 Pa·s). The conductivity of ChCl:MA, ChCl:IA and ChCl:GA, as low as 0.02 S/m, corresponded well with their high viscosity (more than 65 Pa·s). A clear inverse correlation between the conductivity of a DES and its viscosity was basically observed [12,47].

DESs tended to have a high surface tension. From Fig. 2e, the surface tension of ChCl:U surprisingly achieved 84 mN/m, and the surface tension of the other DESs fell between 25 and 70 mN/m, below that of water (72 mN/m) to different degree. The surface tension of a DES is influenced by the intermolecular forces and influences mass transfer at the liquid-liquid or liquid-solid interface [43], therefore affecting the characteristics of biomass processing.

3.1.2. Kamlet-Taft solvatochromic parameters

K-T solvatochromic parameters are used to quantify the hydrogen-bond donating ability (α), the hydrogen bond accepting ability (β) and the polarizability (π*) of DESs [48]. Here we introduced the difference between α and β (α-β) to express the net hydrogen-bond donating ability of DESs [39,49]. The K-T parameters of the prepared DESs and the acidity coefficients (pKa) of the HBDs [50–52] are summarized in Table 2.

The α values of the DESs ranged from 0.904 to 1.343. For different HBDs, the α values increased in the subsequent order: MU < PDO = BDO < U < FM < EG < Gly < GA < LevA < AA < PA < LA:Gly < FA < LA = MA. Amide-based DESs had α values no more than 0.97 and ChCl:MU gave the lowest α. Among the polyol-based DESs, ChCl:Gly displayed the highest α value of 1.056, owing to its maximum number of hydroxyl groups [53]. Acid-based DESs possessed apparently higher α values compared with amide-based and polyol-based DESs, and ChCl:LA and ChCl:MA exhibited the highest α value (1.343) among all the tested solvents. DESs with α above 1.22 were reported to be more acidic resulting from the existence of carboxylic acids in solvents, indicating their ability to liberate H^+ more readily [54]. It was observed that the α value declined slightly with the extension of the alkyl chain length of organic acids. This can be attributed to the fact that the acidity of organic acids decreased with a longer alkyl chain [55].

The β values of the DESs covered a wide range from 0.049 (ChCl:FA) to 1.108 (ChCl:U). The β values increased in the order as follows by different HBDs: FA < MA < PA < LA < GA < LA:Gly < AA < LevA < Gly < BDO < FM < PDO < MU < EG < U. Amide-based DESs had the highest β values (over 1.06), suggesting more sites available for hydrogen bonding. The β values of polyol-based DESs ranged from 1.103 (ChCl:EG) to 0.986 (ChCl:Gly), and decreased with the longer alkyl chain length and more -OH groups in the HBDs. Compared with the amide-based and polyol-based types, acid-based DESs had relatively low β values (no more than 0.74), which showed a trend opposite to their performance on the α values. The α and β values for the same DESs reflected their inclination in hydrogen bonding from different perspectives.

The π* values of the DESs fell between 0.673 and 1.033. For different HBDs, the π* values increased in an order similar to the α values. Overall, acid-based DESs had higher π* values than polyol-based and amide-based DESs, suggesting the close relationship between the polarity of the DESs and the acidity of the HBDs.

α-β denoted the net acidity of DESs [37,39]. The α-β values increased in the following order by the HBDs: MU < U < PDO < BDO < EG < FM < Gly < LevA < AA < GA < LA:Gly < LA < PA < MA < FA. Clearly, acid-based DESs displayed much stronger net hydrogen bond donating ability than amide-based and polyol-based DESs. The net acidity of acid-based DESs (greater than 0.51) far exceeded the highest α-β value

Table 2
Kamlet-Taft parameters of DESs and the pKa values of HBDs.

DES		π*	α	β	α-β	pKa
Amide	ChCl:U	0.710	0.949	1.108	−0.159	26.0 [50]
	ChCl:MU	0.673	0.904	1.102	−0.198	-
	ChCl:FM	0.726	0.968	1.067	−0.099	16.50 [51]
Polyol	ChCl:PDO	0.705	0.942	1.090	−0.148	14.46 [51]
	ChCl:BDO	0.705	0.942	1.066	−0.124	14.83 [51]
	ChCl:Gly	0.798	1.056	0.986	0.071	14.15 [51]
	ChCl:EG	0.736	0.981	1.103	−0.122	14.77 [52]
Acid	ChCl:AA	0.985	1.285	0.703	0.583	4.76 [50]
	ChCl:LevA	0.956	1.250	0.735	0.515	4.65 [51]
	ChCl:GA	0.937	1.227	0.634	0.593	4.34 [52]
	ChCl:LA	1.033	1.343	0.576	0.767	3.86 [52]
	ChCl:PA	0.995	1.297	0.391	0.906	4.86 [51]
	ChCl:MA	1.033	1.343	0.244	1.099	3.40 [52]
	ChCl:FA	1.014	1.320	0.049	1.271	3.77 [52]
	ChCl:IA	-	-	-	-	-
Ternary	ChCl:FM:$FeCl_3$	-	-	-	-	-
	ChCl:Gly:$FeCl_3$	-	-	-	-	-
	ChCl:LA:$FeCl_3$	-	-	-	-	-
	ChCl:LA:Gly	1.004	1.309	0.681	0.627	-

for all amide-based and polyol-based DESs (ChCl:Gly 0.071). In addition, it was reported that the pKa value of an HBD implied the acid strength of a DES [50–52]. Both the α-β and pKa data in Table 2 may thus explain the strong net hydrogen bond donating capacity of acid-based DESs especially the DESs with negative pH such as ChCl:FA and ChCl:MA. It was noted that moderate acid strength was associated with high lignin extraction and great glucose yield [56], while higher acid strength limited the enzymatic digestibility and might induce the charring of the biomass [55]. An appropriate acidity is deemed important for a DES to balance its different effects on lignocellulosic components in pretreatment.

3.2. DES-induced compositional changes

3.2.1. Carbohydrate composition

The composition of DES pretreated biomass is illustrated in Fig. 3. Glucan, xylan and arabinan constituted the main carbohydrate composition. Pretreatment by acidic DESs produced an overall higher glucan fraction (more than 47 %) compared with pretreatment by basic or neutral DESs. ChCl:IA and ChCl:FA caused the highest glucan content of more than 70 %, followed by ChCl:MA (67.3 %) and ChCl:GA (64.3 %). ChCl:MU generated the highest glucan content of 45.1 % among amide-based DESs, and ChCl:BDO 42.5 % among polyol-based DESs. The higher glucan content was directly related to the greater removal of hemicellulose and lignin as well as other DES dissolved components [57,58].

In terms of hemicellulose retention, pretreatment by amide-based and polyol-based DESs maintained a higher xylan fraction (over 22 %) compared with acid-based DES pretreatment (less than 19.5 %). Especially, ChCl:U, ChCl:MU, ChCl:PDO and ChCl:BDO generated a xylan content of more than 25 %, while the xylan content in ChCl:FA pretreated biomass even decreased to 9.6 %. In addition, acid-based DESs removed almost all arabinan, while amide-based and polyol-based DESs kept an arabinan content of up to 3.2 %. The compositional change of xylan and arabinan reflected the deconstruction of the hemicellulose structures, and was in the meanwhile affected by the removal of lignin and other components. The significant decrease in xylan and arabinan content caused by acid-based DESs agreed with the previous reports that acidic DESs were advantageous for hemicellulose solubilization and a high acid proportion in DESs could remarkably degrade hemicellulose [34,59].

For ternary DESs, the introduction of $FeCl_3$ into the HBDs caused considerable compositional changes compared with the corresponding binary DESs. The addition of $FeCl_3$ to ChCl:Gly and ChCl:LA resulted in an increase in the glucan content and a decrease in xylan content, which was the opposite effect of the addition of $FeCl_3$ to ChCl:FM. The significant effect of the addition of $FeCl_3$ in DESs on the biomass composition was previously reported [29,30]. Interestingly, the ternary DES ChCl:LA:Gly, whose acidity was enhanced by the addition of lactic acid, resulted in an intermediate level between the compositional data caused by ChCl:Gly and ChCl:LA, seemingly combining the pretreatment effects of the two binary DESs.

3.2.2. Delignification and solid yield

Fig. 4 shows the removal of lignin and the solid yields of the DES-pretreated biomass. Basically, acid-based DESs (pH < 1.7) caused delignification to a greater degree than amide-based and polyol-based DESs (4.4 < pH < 10.0). ChCl:FA obtained the highest delignification rate of 80.7 %, followed by ChCl:IA (80.5 %) and ChCl:LA (78.1 %). In contrast, delignification by amide-based and polyol-based DESs was less than 37 %. The lignin removal rate of ChCl:PDO and ChCl:BDO pretreatment was just 21 %. The mechanisms of lignin removal by different DESs are similar and mainly based on the effect of hydrogen bonding. The HBD and the HBA in an acidic DES can form stable hydrogen bonds and generate active proton pairs to dissolve and extract biomass components. For a neutral DES, hydroxyl groups in polyols can interact with free and etherified hydroxyl groups in lignin [23]. The HBDs of a basic DES contained the major functional groups -C═O and -N-H, which formed strong hydrogen bonds with lignin-dominated regions in lignocellulose [60]. Although the functional groups that interact with lignin are not the same for different DESs, they do induce varying degrees of cracking and solubilization of the lignin structure. While acidic DESs may be good solvents for lignin extraction [56,61], it was also noted that lignin recovered by alkaline DESs possessed well-preserved structures, which elevates the utilization value of the lignocellulosic component [24].

It was found that delignification by ChCl:FA (C_1) was higher than by ChCl:AA (C_2) and ChCl:PA (C_3). It is reported that the length of the aliphatic chain and the number of functional groups of HBDs in acidic DESs significantly affected the extraction of lignin [62,63]. A decreased aliphatic chain length would promote the proton donating ability of DESs. With the same aliphatic chain length, the extra -OH functional group in ChCl:LA led to increased lignin removal in comparison with ChCl:PA. IA, LevA and GA shared similar aliphatic chain lengths (C_5)

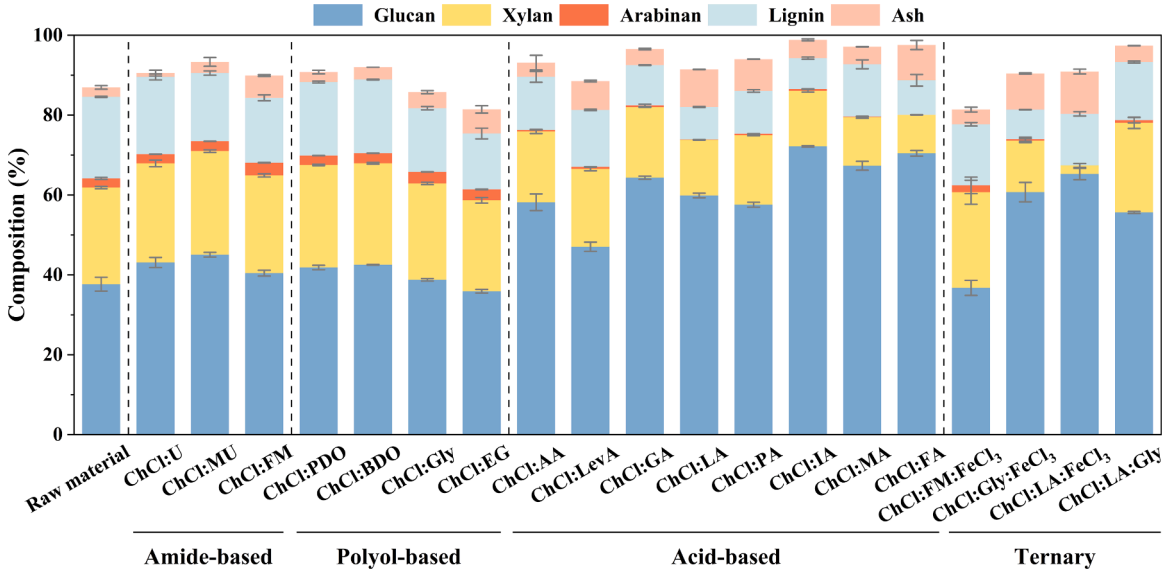


Fig. 3. Chemical composition of raw biomass and DES-pretreated biomass.

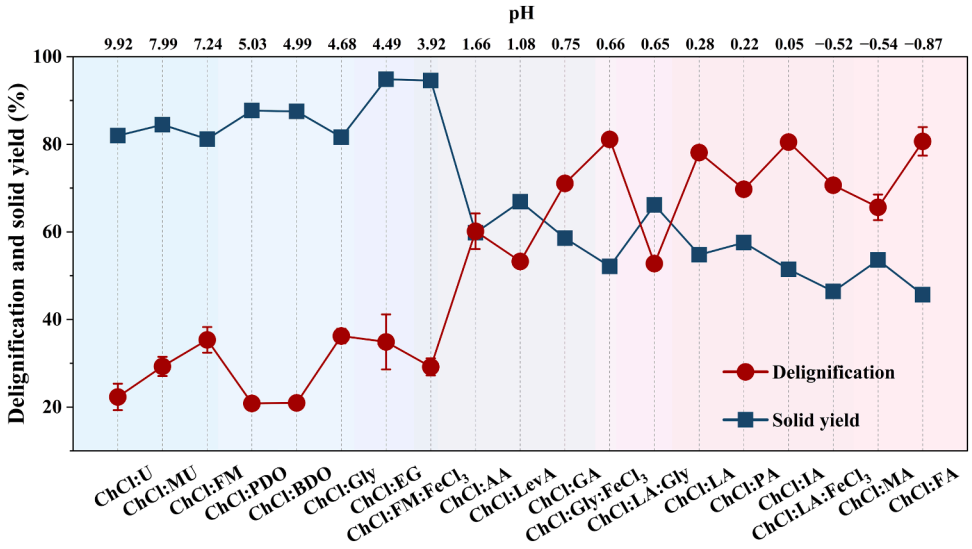


Fig. 4. Delignification and solid yield of DES-pretreated biomass.

with different functional groups, and delignification increased in the order of ChCl:LevA < ChCl:GA < ChCl:IA. The extra -COOH groups in GA provided acidity and allowed for more hydrogen bond formation with the HBA, facilitating the removal of lignin, which differed from the C═O group in LevA. The C═C bond in IA improved the ability of delignification compared with GA. It was reported that the functional group C═C in the HBD can affect the ability of lignin dissolution [64]. Moreover, comparing MA (additional -COOH group) with LA (longer aliphatic chain), ChCl:MA performed better in delignification. However, a longer alkyl chain and an increased number of functional groups in HBDs would increase viscosity and restrict mobility, which may be detrimental to the use of DESs.

The introduction of $FeCl_3$ leading to a decrease in pH exerted different impacts on lignin removal. Compared with the corresponding binary DES, delignification by the ternary DES ChCl:Gly:$FeCl_3$ substantially increased to 81.1 %, more than twice that by ChCl:Gly (36.2 %), whereas ChCl:FM:$FeCl_3$ and ChCl:LA:$FeCl_3$ showed no improvement in delignification. It is inferred that the acidic protons could greatly enhance the delignification capacity of polyol-based DESs. The addition of Lewis acid was thought to improve the hydrogen bond acceptance and elevate the content of acidic sites in a DES, thereby efficiently cleaving the β-O-4 linkages in lignin [65,66]. Also, metal chloride salts (Cl^-) can provide active protons for the reactions and form stronger hydrogen bonds, facilitating the cleavage of ether bonds in lignin-carbohydrate complexes [32]. In addition, the ternary DES ChCl:LA:Gly caused delignification falling between the effects of ChCl:LA and ChCl:Gly, which is interesting.

The solid yield represents the percentage of undissolved biomass subjected to DES treatment. Amide-based and polyol-based DESs exhibited a high solid yield of more than 80 %, with ChCl:EG reaching approximately 95 %, which was consistent with the result of the previous study [8]. Acid-based DESs possessed a relatively low solid yield below 67 %, with only 45.7 % for ChCl:FA. The solid yield generally decreased with the increasing acidity of the DESs [34], showing a tendency opposite to delignification.

3.2.3. *Relationship between compositional changes*

The relationship between the glucan content and the removal of lignin and xylan in biomass pretreated by different DESs is presented in Fig. 5. The glucan content correlated positively with delignification (R^2 = 0.82) and xylan removal (R^2 = 0.85). Meanwhile, delignification and xylan removal exhibited a positive relationship with R^2 = 0.90, which was also noted before [59]. DESs can create new hydrogen bonds between DES molecules and lignocellulosic components, thus breaking intermolecular hydrogen bonding in biomass and destroying the structural recalcitrance. Especially in non-crystalline zones, mostly composed of lignin and xylan, lignocellulosic structures are looser and more susceptible to exogenous attack compared with those in crystalline zones (cellulose). Consequently, when lignin was removed by dissolving in a DES, xylan structures were also easily dismantled and removed. And possibly vice versa. The removal of lignin and xylan thus displayed positive correlation to a considerable degree. Additionally, it is easily

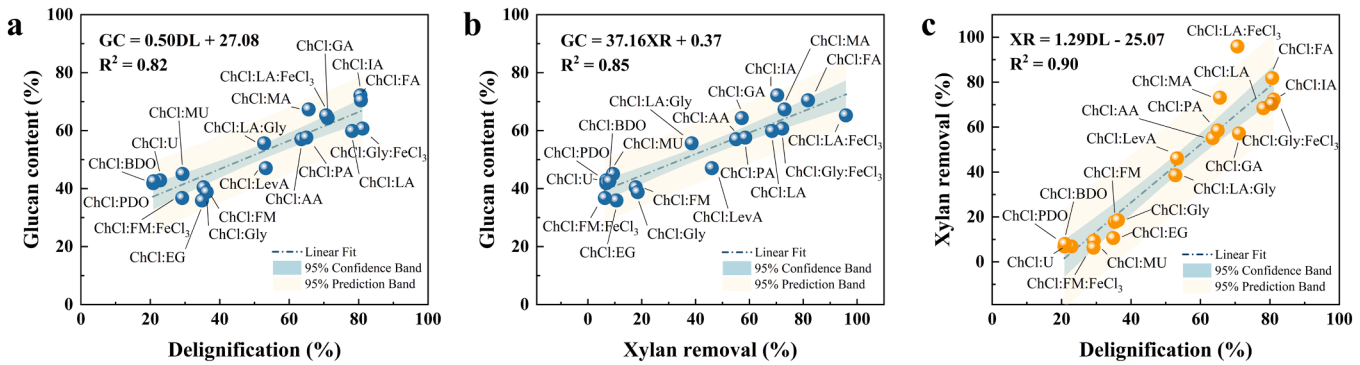


Fig. 5. Correlation between the glucan content (GC), delignification (DL) and xylan removal (XR) in pretreated biomass. (a) glucan content vs xylan removal, (b) glucan content vs delignification, (c) xylan removal and delignification.

understood that the synergistic removal of lignin and hemicellulose would result in an increase in the glucan content in biomass pretreated by the same DES [67]. The fact that the above relationship held for biomass pretreated by various kinds of DESs was somewhat significant. This indicated that various ChCl-based DESs basically followed the same mechanism when interacting with biomass, which was primarily responsible for efficient cleavage of lignin-carbohydrate linkages and good preservation of cellulose.

3.3. Relationship between Kamlet-Taft parameters and compositional changes

Kamlet-Taft parameters represent the properties of molecular bonding in DESs which can be closely related to the pretreatment. Fig. 6 exhibited the correlation between Kamlet-Taft parameters and compositional characteristics of the pretreated biomass.

The glucan content (Figs. 6a-6c), delignification (Figs. 6d-6f) and xylan removal (Figs. 6g-6i) showed similar relationships with K-T parameters. The glucan content, delignification and xylan removal were

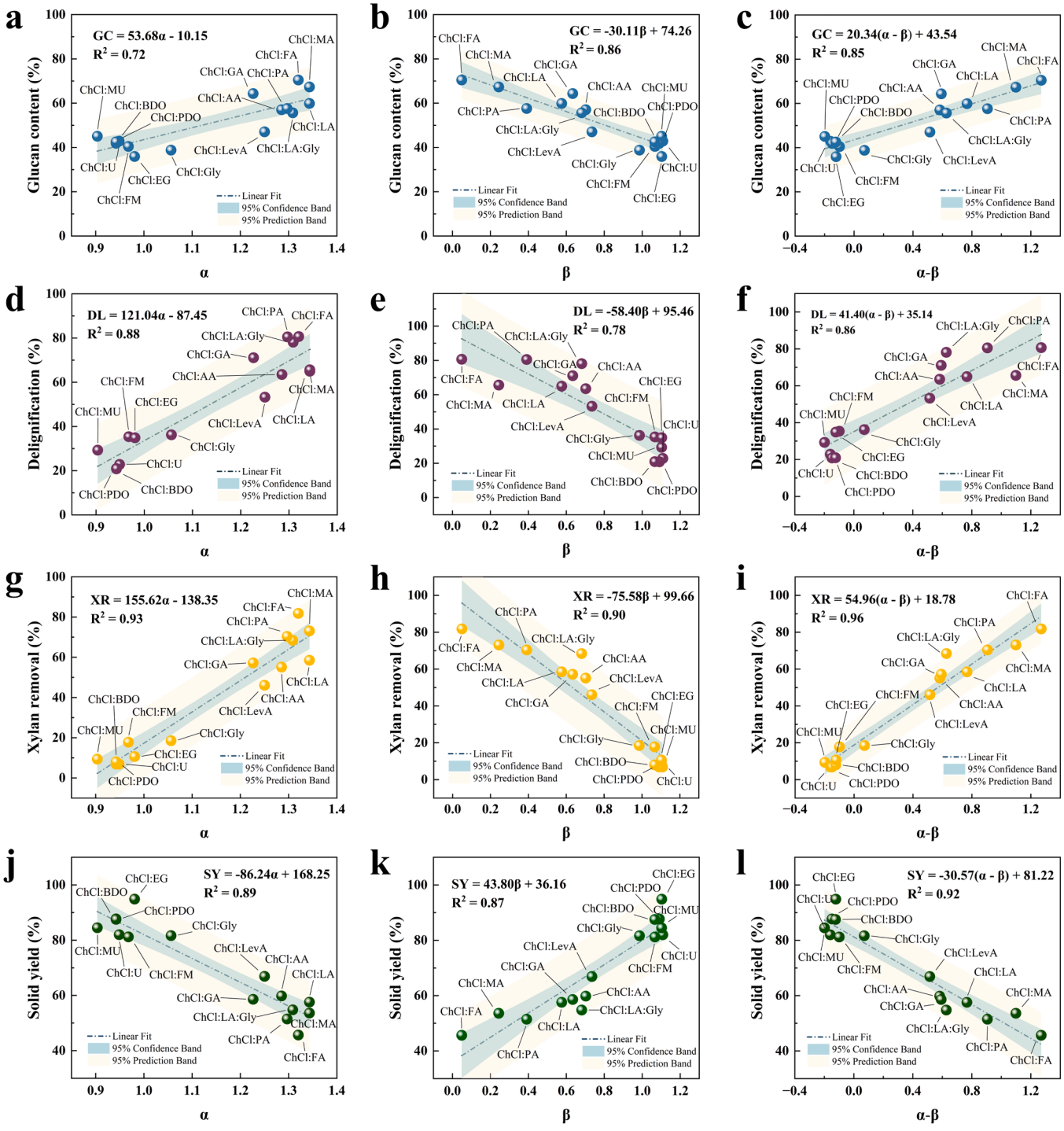


Fig. 6. Correlation between Kamlet-Taft parameters (α, β and α-β) and compositional characteristics of pretreated biomass. (a-c) glucan content (GC), (d-f) delignification (DL), (g-i) xylan removal (XR), (j-l) solid yield (SY).

positively correlated with the α value (R^2 = 0.72, 0.88 and 0.93, respectively) and α-β value (R^2 = 0.85, 0.86 and 0.96, respectively), and negatively correlated with the β value (R^2 = 0.86, 0.78 and 0.90, respectively). Exhibiting an opposite trend, the solid yield decreased with elevated α value (R^2 = 0.89) and α-β value (R^2 = 0.92), and increased with elevated β value (R^2 = 0.87). It was reported that a higher α value was favorable for better glucan retention and delignification [68,69], and a higher β value was disadvantageous to the removal of lignin and xylan. More importantly, the α-β value reflecting the net hydrogen-bond donating capacity of DESs showed a clearly positive relationship with lignin solubility and xylan removal, as also noticed in other work [53,64]. The alteration of solid yield was predominately attributed to the change in hemicellulose fractions and lignin content, which was in the end induced by the acidic properties of DESs. DESs with high α-β have high acidic strength, can form strong hydrogen bonds with lignin, and tend to cleave ether bonds, glycosidic bonds and lignin-carbohydrate linkages, especially the β-O-4 bonds in lignocellulose [65,70]. The α-β value may thus be taken as an indicator to predict the ability of lignin and xylan removal of DESs, signaling the potential of pretreatment to a great degree.

3.4. Enzymatic hydrolysis of pretreated biomass

The enzymatic digestibility of DES pretreated biomass is displayed in Fig. 7. It is seen that enzymatic hydrolysis was heavily dependent on the HBD type of the DESs. Pretreatment by amide-based (Fig. 7a) and polyol-based DESs (Fig. 7b) enhanced enzymatic digestibility to a modest degree, with the highest glucose yield of 38.5 % for ChCl:FM among amide-based DESs, and 46.7 % for ChCl:Gly among polyol-based DESs. Acid-based DESs varied in their effects on enzymatic hydrolysis. The highest glucose yield of 65.3 % was obtained for ChCl:LA, followed by 58.8 % for ChCl:LevA and 45.7 % for ChCl:GA. The corresponding xylose yields reached 37.8 %, 28.9 % and 35.6 %, respectively. Amide-based DESs, having a higher alkalinity, exhibited a similar effect on enzymatic hydrolysis to polyol-based DESs, and most acid-based DESs had better performance in saccharification than amide-based and polyol-based DESs [71]. The higher saccharification for acid-based DESs should be related to the greater removal of lignin and hemicellulose, though lignin deposition might have a negative impact on enzymatic hydrolysis [23].

Based on the enzymatic hydrolysis performance, three HBDs, i.e. FM from amides, Gly from polyols and LA from acids, were chosen to synthesize ternary DESs by simultaneously incorporating the Lewis acid $FeCl_3$. ChCl:LA:Gly was synthesized considering good enzymatic hydrolysis performance resulted from the treatment of ChCl:LA and ChCl:Gly. As shown in Fig. 7d, the introduction of $FeCl_3$ generally decreased the sugar yields by enzymatic hydrolysis, which was inconsistent with previous report [30,32], possibly due to different DES preparation and pretreatment conditions. Surprisingly, ChCl:LA:Gly produced outstanding enzymatic hydrolysis performance among all the DESs prepared, with a glucose yield of 78.2 %, nearly 5 folds of that of raw biomass (16.1 %), and a xylose yield of 44.9 %. It was reported that the glucose yield was about 70 % for corn straw pretreated by ChCl:LA:Gly (100 ℃ for 6 h) at a high solid-to-liquid ratio (1:10) [72], which is comparable to the result here. The enhanced delignification and enzymatic hydrolysis by the addition of polyols in acid-based binary DESs have been reported [33,34]. Polyols can selectively remove lignin by optimizing the acidity of the DESs while limiting excessive xylan degradation, thus favoring the recovery of carbohydrates. A ternary DES composed of two HBDs implies more room for functional adjustment in carbohydrate fractionation and lignin structure protection compared with its binary counterparts [73,74]. Ternary ChCl-based DESs may deserve in-depth research on their properties and pretreatment behavior as a green and tunable solvent.

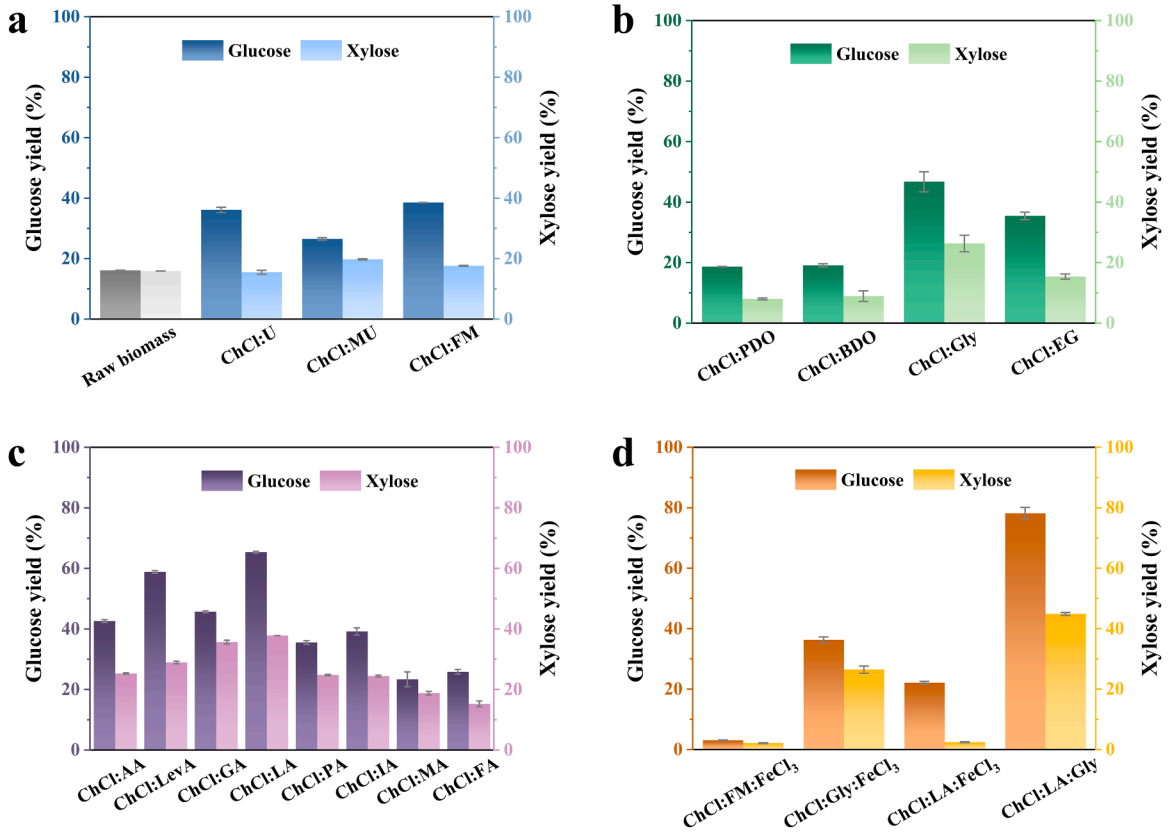


Fig. 7. Glucose and xylose yields by enzymatic digestibility of DES pretreated biomass at 72 h. (a) amide-based DESs, (b) polyol-based DESs, (c) acid-based DESs, and (d) ternary DESs.

4. Conclusion

This study comparatively investigated the pretreatment of wheat straw by nineteen ChCl-based DESs (amide-based, polyol-based, acid-based and ternary DESs) with different physicochemical characteristics. Acid-based DESs, showing strong net acidity, generally outperformed basic and neutral DESs in delignification and hemicellulose removal as well as enzymatic hydrolysis of the pretreated biomass. Removal of lignin and xylan were positively correlated with the K-T parameters α and α-β of the DESs, and negatively with the parameter β, indicating the close relationship between the hydrogen bond donating ability and biomass fractionation. Enzymatic digestibility depended heavily on the HBD type of the DESs. The highest glucose yield of 78.2 % was achieved by ChCl:LA:Gly, implying the potential of a ternary structure for carbohydrate recovery. This work has offered a comparative insight into biomass pretreatment by ChCl-based DESs with different physicochemical properties, which may benefit the design of an appropriate DES.

CRediT authorship contribution statement

Jinwen Guo: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Guoce Yu:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Jianlong Wang:** Writing – review & editing, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the State Administration of Science, Technology and Industry for National Defence, China ([2017]1474) and the Foshan-Tsinghua Innovation Special Fund (FTISF) (2019THFS0133).

Data availability

Data will be made available on request.

References

[1] A.J. Ragauskas, C.K. Williams, B.H. Davison, G. Britovsek, J. Cairney, C.A. Eckert, W.J. Frederick, J.P. Hallett, D.J. Leak, C.L. Liotta, J.R. Mielenz, R. Murphy, R. Templer, T. Tschaplinski, The Path Forward for Biofuels and Biomaterials, Science 311 (5760) (2006) 484–489.
[2] W. Wang, D. Lee, Lignocellulosic biomass pretreatment by deep eutectic solvents on lignin extraction and saccharification enhancement: A review, Bioresour. Technol. 339 (2021) 125587.
[3] Z. Chen, L. Chen, K.S. Khoo, V.K. Gupta, M. Sharma, P.L. Show, P. Yap, Exploitation of lignocellulosic-based biomass biorefinery: A critical review of renewable bioresource, sustainability and economic views, Biotechnol. Adv. 69 (2023) 108265.
[4] C.G. Yoo, X. Meng, Y. Pu, A.J. Ragauskas, The critical role of lignin in lignocellulosic biomass conversion and recent pretreatment strategies: a comprehensive review, Bioresour. Technol. 301 (2020) 122784.
[5] R. Li, H. Ruan, D. Zhang, C. Zhu, C. Lai, Q. Yong, Tween 80 reversed adverse effects of combined autohydrolysis and p-toluenesulfonic acid pretreatment on enzymatic hydrolysis of poplar, Bioresour. Technol. 393 (2024) 130056.
[6] F. Wang, D. Ouyang, Z. Zhou, S.J. Page, D. Liu, X. Zhao, Lignocellulosic biomass as sustainable feedstock and materials for power generation and energy storage, J. Energy Chem. 57 (2021) 247–280.
[7] A. Sidana, S.K. Yadav, Recent developments in lignocellulosic biomass pretreatment with a focus on eco-friendly, non-conventional methods, J. Clean. Prod. 335 (2022) 130286.
[8] A. Procentese, E. Johnson, V. Orr, A.G. Campanile, J.A. Wood, A. Marzocchella, L. Rehmann, Deep eutectic solvent pretreatment and subsequent saccharification of corncob, Bioresour. Technol. 192 (2015) 31–36.
[9] S.K. Bhatia, S.S. Jagtap, A.A. Bedekar, R.K. Bhatia, A.K. Patel, D. Pant, J. Rajesh Banu, C.V. Rao, Y. Kim, Y. Yang, Recent developments in pretreatment technologies on lignocellulosic biomass: Effect of key parameters, technological improvements, and challenges, Bioresour. Technol. 300 (2020) 122724.
[10] B. Kumar, N. Bhardwaj, K. Agrawal, V. Chaturvedi, P. Verma, Current perspective on pretreatment technologies using lignocellulosic biomass: an emerging biorefinery concept, Fuel Process. Technol. 199 (2020) 106244.
[11] A.P. Abbott, D. Boothby, G. Capper, D.L. Davies, R.K. Rasheed, Deep eutectic solvents formed between choline chloride and carboxylic acids: versatile alternatives to ionic liquids, J. Am. Chem. Soc. 126 (29) (2004) 9142–9147.
[12] E.L. Smith, A.P. Abbott, K.S. Ryder, Deep eutectic solvents (DESs) and their applications, Chem. Rev. 114 (21) (2014) 11060–11082.
[13] M. Francisco, A. van den Bruinhorst, M.C. Kroon, New natural and renewable low transition temperature mixtures (LTTMs): screening as solvents for lignocellulosic biomass processing, Green. Chem. 14 (8) (2012) 2153.
[14] Y. Wang, Q. Liu, C. Yan, G. Song, W.S. Price, G. Zheng, A.M. Torres, Z. Xue, Deep eutectic solvent-driven mild lignocellulose pretreatment: Unlocking lignin valorization and carbohydrate digestibility, Chem. Eng. J. 504 (2025) 158825.
[15] Y. Bao, Y. Wang, C. Yan, Z. Xue, Deep eutectic solvents for fractionation and valorization of lignocellulose, Green. Chem. Eng. 6 (1) (2025) 21–35.
[16] B.B. Hansen, S. Spittle, B. Chen, D. Poe, Y. Zhang, J.M. Klein, A. Horton, L. Adhikari, T. Zelovich, B.W. Doherty, B. Gurkan, E.J. Maginn, A. Ragauskas, M. Dadmun, T.A. Zawodzinski, G.A. Baker, M.E. Tuckerman, R.F. Savinell, J. R. Sangoro, Deep eutectic solvents: a review of fundamentals and applications, Chem. Rev. 121 (3) (2021) 1232–1285.
[17] Y.T. Tan, A.S.M. Chua, G.C. Ngoh, Deep eutectic solvent for lignocellulosic biomass fractionation and the subsequent conversion to bio-based products - a review, Bioresour. Technol. 297 (2020) 122522.
[18] D. Haldar, M.K. Purkait, A review on the environment-friendly emerging techniques for pretreatment of lignocellulosic biomass: mechanistic insight and advancements, Chemosphere 264 (2021) 128523.
[19] S. Massayev, K.M. Lee, Evaluation of deep eutectic solvent pretreatment towards efficacy of enzymatic saccharification using multivariate analysis techniques, J. Clean. Prod. 360 (2022) 132239.
[20] T. Suopajarvi, J.A. Sirvio, H. Liimatainen, Nanofibrillation of deep eutectic solvent-treated paper and board cellulose pulps, Carbohyd. Polym. 169 (2017) 167–175.
[21] J.A. Sirviö, K. Hyypiö, S. Asaadi, K. Junka, H. Liimatainen, High-strength cellulose nanofibers producedvia swelling pretreatment based on a choline chloride–imidazole deep eutectic solvent, Green. Chem. 22 (5) (2020) 1763–1775.
[22] Y. Guo, L. Xu, F. Shen, J. Hu, M. Huang, J. He, Y. Zhang, S. Deng, Q. Li, D. Tian, Insights into lignocellulosic waste fractionation for lignin nanospheres fabrication using acidic/alkaline deep eutectic solvents, Chemosphere 286 (2022) 131798.
[23] C. Zhang, S. Xia, P. Ma, Facile pretreatment of lignocellulosic biomass using deep eutectic solvents, Bioresour. Technol. 219 (2016) 1–5.
[24] K.H. Kim, Y. Mottiar, K. Jeong, P.H.N. Tran, N.T. Tran, J. Zhuang, C.S. Kim, H. Lee, G. Gong, J.K. Ko, S. Lee, S.Y. Kim, J.Y. Shin, H. Jeong, H.K. Song, C.G. Yoo, N. Kim, S.D. Mansfield, One-pot conversion of engineered poplar into biochemicals and biofuels using biocompatible deep eutectic solvents, Green. Chem. 24 (23) (2022) 955–968.
[25] Y. Yu, W. Cheng, Y. Li, T. Wang, Q. Xia, Y. Liu, H. Yu, Tailored one-pot lignocellulose fractionation to maximize biorefinery toward versatile xylochemicals and nanomaterials, Green. Chem. 24 (8) (2022) 3257–3268.
[26] G. Xu, J. Ding, R. Han, J. Dong, Y. Ni, Enhancing cellulose accessibility of corn stover by deep eutectic solvent pretreatment for butanol fermentation, Bioresour. Technol. 203 (2016) 364–369.
[27] Y. Liu, W. Chen, Q. Xia, B. Guo, Q. Wang, S. Liu, Y. Liu, J. Li, H. Yu, Efficient Cleavage of Lignin-Carbohydrate Complexes and Ultrafast Extraction of Lignin Oligomers from Wood Biomass by Microwave-Assisted Treatment with Deep Eutectic Solvent, Chemsuschem 10 (8) (2017) 1692–1700.
[28] J. Cheng, C. Huang, Y. Zhan, S. Han, J. Wang, X. Meng, C.G. Yoo, G. Fang, A. J. Ragauskas, Effective biomass fractionation and lignin stabilization using a diol DES system, Chem. Eng. J. 443 (2022) 136395.
[29] D. Zhang, J. Liu, H. Xu, H. Liu, Y. He, Improving saccharification efficiency of corn stover through ferric chloride-deep eutectic solvent pretreatment, Bioresour. Technol. 399 (2024) 130579.
[30] C. He, F. Shen, D. Tian, M. Huang, L. Zhao, Q. Yu, F. Shen, Lewis acid/base mediated deep eutectic solvents intensify lignocellulose fractionation to facilitate enzymatic hydrolysis and lignin nanosphere preparation, Int. J. Biol. Macromol. 254 (2024) 127853.
[31] Z. Wang, H. Li, X. Lin, L. Tang, J. Chen, J. Mo, R. Yu, X. Shen, Novel recyclable deep eutectic solvent boost biomass pretreatment for enzymatic hydrolysis, Bioresour. Technol. 307 (2020) 123237.
[32] Y. Feng, T.L. Eberhardt, F. Meng, C. Xu, H. Pan, Efficient extraction of lignin from moso bamboo by microwave-assisted ternary deep eutectic solvent pretreatment for enhanced enzymatic hydrolysis, Bioresour. Technol. 400 (2024) 130666.
[33] X. Yue, J. Lin, T. Suopajärvi, O. Mankinen, A. Mikkelson, R. Liu, H. Huttunen, L. Chen, C. Xu, V. Telkki, S. Sun, H. Liimatainen, Conversion of highly polymerized lignin into monophenolic products via pyrolysis: A comparative study of acidic and alkaline depolymerization pretreatments using deep eutectic solvents, Chem. Eng. J. 478 (2023) 147368.
[34] L. Zhang, Y. Song, Q. Wang, X. Zhang, Culturing Rhodotorula glutinis in fermentation-friendly deep eutectic solvent extraction liquor of lignin for producing microbial lipid, Bioresour. Technol. 337 (2021) 125475.
[35] H. Xu, Y. Kong, J. Peng, X. Song, X. Che, S. Liu, W. Tian, Multivariate analysis of the process of deep eutectic solvent pretreatment of lignocellulosic biomass, Ind. Crop. Prod. 150 (2020) 112363.

[36] H. Xu, Y. Kong, J. Peng, X. Song, Y. Liu, Z. Su, B. Li, C. Gao, W. Tian, Comprehensive analysis of important parameters of choline chloride-based deep eutectic solvent pretreatment of lignocellulosic biomass, Bioresour. Technol. 319 (2021) 124209.
[37] Q. Liu, X. Zhao, D. Yu, H. Yu, Y. Zhang, Z. Xue, T. Mu, Novel deep eutectic solvents with different functional groups towards highly efficient dissolution of lignin, Green. Chem. 21 (19) (2019) 5291–5297.
[38] H. Yu, Z. Xue, R. Shi, F. Zhou, T. Mu, Lignin dissolution and lignocellulose pretreatment by carboxylic acid based deep eutectic solvents, Ind. Crop. Prod. 184 (2022) 115049.
[39] Y. Zhang, H. Ren, B. Li, S.M. Udin, H. Maarof, W. Zhou, F. Cheng, J. Yang, Y. Liu, H. Alias, E. Duan, Mechanistic insights into the lignin dissolution behavior in amino acid based deep eutectic solvents, Int. J. Biol. Macromol. 242 (2023) 124829.
[40] A. Sluiter, B. Hames, R. Ruiz, C. Scarlata, J. Sluiter, D. Templeton, D. Crocker, Determination of structural carbohydrates and lignin in biomass, in: laboratory analytical procedure (LAP), Natl. Renew. Energy Lab. Denver USA (2008).
[41] A. Satlewal, R. Agrawal, S. Bhagia, J. Sangoro, A.J. Ragauskas, Natural deep eutectic solvents for lignocellulosic biomass pretreatment: recent developments, challenges and novel opportunities, Biotechnol. Adv. 36 (8) (2018) 2032–2050.
[42] Q. Zhang, K. De Oliveira Vigier, S. Royer, F. Jérôme, Deep eutectic solvents: syntheses, properties and applications, Chem. Soc. Rev. 41 (21) (2012) 7108–7146.
[43] R.K. Ibrahim, M. Hayyan, M.A. AlSaadi, S. Ibrahim, A. Hayyan, M.A. Hashim, Physical properties of ethylene glycol-based deep eutectic solvents, J. Mol. Liq. 276 (2019) 794–800.
[44] M.S. Álvarez, M.A. Longo, F.J. Deive, A. Rodríguez, Synthesis and characterization of a lipase-friendly DES based on cholinium dihydrogen phosphate, J. Mol. Liq. 340 (2021) 117230.
[45] D. Shi, F. Zhou, W. Mu, C. Ling, T. Mu, G. Yu, R. Li, Deep insights into the viscosity of deep eutectic solvents by an XGBoost-based model plus SHapley Additive exPlanation, Phys. Chem. Chem. Phys. 24 (42) (2022) 26029–26036.
[46] X. Li, C. Ning, L. Li, W. Liu, Q. Ren, Q. Hou, Fabricating lignin-containing cellulose nanofibrils with unique properties from agricultural residues with assistance of deep eutectic solvents, Carbohyd. Polym. 274 (2021) 118650.
[47] I.M. Aroso, A. Paiva, R.L. Reis, A.R.C. Duarte, Natural deep eutectic solvents from choline chloride and betaine - physicochemical properties, J. Mol. Liq. 241 (2017) 654–661.
[48] Á. Lobato-Rodríguez, B. Gullón, A. Romaní, P. Ferreira-Santos, G. Garrote, P. G. Del-Río, Recent advances in biorefineries based on lignin extraction using deep eutectic solvents: a review, Bioresour. Technol. 388 (2023) 129744.
[49] Y. Zhang, H. Ren, B. Li, S.M. Udin, H. Maarof, W. Zhou, F. Cheng, J. Yang, Y. Liu, H. Alias, E. Duan, Mechanistic insights into the lignin dissolution behavior in amino acid based deep eutectic solvents, Int. J. Biol. Macromol. 242 (2023) 124829.
[50] H. Xu, J. Peng, Y. Kong, Y. Liu, Z. Su, B. Li, X. Song, S. Liu, W. Tian, Key process parameters for deep eutectic solvents pretreatment of lignocellulosic biomass materials: A review, Bioresour. Technol. 310 (2020) 123416.
[51] D.A. Evans, D.H. Ripin, A pKa values compilation. ⟨https://organicchemistrydata.org/hansreich/resources/pka/⟩, 2024 (accessed 9 July 2025).
[52] R. Williams, A pKa values in water compilation. ⟨https://organicchemistrydata.org/hansreich/resources/pka/⟩, 2022 (accessed 9 July 2025).
[53] L. Zhang, H. Yu, S. Liu, Y. Wang, T. Mu, Z. Xue, Kamlet–Taft parameters of deep eutectic solvents and their relationship with dissolution of main lignocellulosic components, Ind. Eng. Chem. Res. 62 (29) (2023) 11723–11734.
[54] Q. Xia, Y. Liu, J. Meng, W. Cheng, W. Chen, S. Liu, Y. Liu, J. Li, H. Yu, Multiple hydrogen bond coordination in three-constituent deep eutectic solvents enhances lignin fractionation from biomass, Green. Chem. 20 (12) (2018) 2711–2721.
[55] A.R. Mankar, A. Pandey, K.K. Pant, Microwave-assisted extraction of lignin from coconut coir using deep eutectic solvents and its valorization to aromatics, Bioresour. Technol. 345 (2022) 126528.
[56] V. Provost, S. Dumarcay, I. Ziegler-Devin, M. Boltoeva, D. Trebouet, M. Villain-Gambier, Deep eutectic solvent pretreatment of biomass: Influence of hydrogen bond donor and temperature on lignin extraction with high β-O-4 content, Bioresour. Technol. 349 (2022) 126837.
[57] Z. Guo, Z. Ling, C. Wang, X. Zhang, F. Xu, Integration of facile deep eutectic solvents pretreatment for enhanced enzymatic hydrolysis and lignin valorization from industrial xylose residue, Bioresour. Technol. 265 (2018) 334–339.
[58] P. Zhao, F. Pu, C. Su, Y. Wan, T. Huang, X. Hou, D. Cai, Towards valorization of rice straw into bioethanol and lignin: Emphasizing critical role of deep eutectic solvent components in biorefining process, Bioresour. Technol. 399 (2024) 130635.
[59] Y. Oh, S. Park, D. Jung, K.K. Oh, S.H. Lee, Effect of hydrogen bond donor on the choline chloride-based deep eutectic solvent-mediated extraction of lignin from pine wood, Int. J. Biol. Macromol. 165 (2020) 187–197.
[60] T. Xiao, M. Hou, X. Guo, X. Cao, C. Li, Q. Zhang, W. Jia, Y. Sun, Y. Guo, H. Shi, Recent progress in deep eutectic solvent(DES) fractionation of lignocellulosic components: A review, Renew. Sustain. Energy Rev. 192 (2024) 114243.
[61] K. Kohli, S. Katuwal, A. Biswas, B.K. Sharma, Effective delignification of lignocellulosic biomass by microwave assisted deep eutectic solvents, Bioresour. Technol. 303 (2020) 122897.
[62] X. Liang, Y. Zhu, B. Qi, S. Li, J. Luo, Y. Wan, Structure-property-performance relationships of lactic acid-based deep eutectic solvents with different hydrogen bond acceptors for corn stover pretreatment, Bioresour. Technol. 336 (2021) 125312.
[63] Y.T. Tan, G.C. Ngoh, A.S.M. Chua, Effect of functional groups in acid constituent of deep eutectic solvent for extraction of reactive lignin, Bioresour. Technol. 281 (2019) 359–366.
[64] R. Kandanelli, C. Thulluri, R. Mangala, P.V.C. Rao, S. Gandham, H.R. Velankar, A novel ternary combination of deep eutectic solvent-alcohol (DES-OL) system for synergistic and efficient delignification of biomass, Bioresour. Technol. 265 (2018) 573–576.
[65] Y. Bai, X. Zhang, Z. Wang, T. Zheng, J. Yao, Deep eutectic solvent with bifunctional Bronsted-Lewis acids for highly efficient lignocellulose fractionation, Bioresour. Technol. 347 (126723) (2022) 126723.
[66] Z. Wang, S. Hong, J. Wen, C. Ma, L. Tang, H. Jiang, J. Chen, S. Li, X. Shen, T. Yuan, Lewis acid-facilitated deep eutectic solvent (DES) pretreatment for producing high-purity and antioxidative lignin, Acs Sustain. Chem. Eng. 8 (2) (2020) 1050–1057.
[67] F. Shen, S. Wu, M. Huang, L. Zhao, J. He, Y. Zhang, S. Deng, J. Hu, D. Tian, F. Shen, Integration of lignin microcapsulated pesticide production into lignocellulose biorefineries through $FeCl_3$-mediated deep eutectic solvent pretreatment, Green. Chem. 24 (13) (2022) 5242–5254.
[68] H. Yu, Z. Xue, Y. Wang, C. Yan, L. Chen, T. Mu, Enabling efficient dissolution and fractionation of lignin by renewable and adjustable dimethyl isosorbide-based solvent systems, Sep. Purif. Technol. 306 (2023) 122688.
[69] Y. Zhong, J. Wu, H. Kang, R. Liu, Choline hydroxide based deep eutectic solvent for dissolving cellulose, Green. Chem. 24 (6) (2022) 2464–2475.
[70] Y. Xie, J. Zhao, P. Wang, Z. Ling, Q. Yong, Natural arginine-based deep eutectic solvents for rapid microwave-assisted pretreatment on crystalline bamboo cellulose with boosting enzymatic conversion performance, Bioresour. Technol. 385 (2023) 129438.
[71] S. Thi, K.M. Lee, Comparison of deep eutectic solvents (DES) on pretreatment of oil palm empty fruit bunch (OPEFB): cellulose digestibility, structural and morphology, Bioresour. Technol. 282 (2019) 525–529.
[72] S. Li, Y. Wang, Q. Dong, Z. Yuan, T. Mu, Z. Xue, L. Cao, Polyol-assisted ternary deep eutectic solvent protective lignocellulose pretreatment for high-efficiency xylan utilization and ethanol production, Carbohyd. Polym. 346 (2024) 122628.
[73] G. Wu, Y. Cheng, C. Huang, C. Yong, Y. Fu, Deep eutectic solvent engineering: a novel ternary system for efficient lignocellulose extraction, Green. Chem. 27 (5) (2025) 1556–1569.
[74] Z. Jia, X. Li, S. Xu, Y. Sun, L. Chen, J. Ye, W. Li, T. Wu, J. Li, S. Wang, X. Xu, Novel ternary deep eutectic solvents pretreatment of corn stalk to realize high-value utilization, Chem. Eng. J. 500 (2024) 156567.